

Henry A. Odunze

BASc. Electrical Engineering

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Skills

Automation & Quality	Selenium, API/UI testing, E2E frameworks, SpecFlow, RCA, shift-left quality
AI-Driven Engineering	LLM workflows, agentic tooling, Claude Code, MCP, LLM evaluation frameworks
CI/CD & Delivery	GitHub Actions, Jenkins, CircleCI, Azure DevOps, ArgoCD, quality gates
Cloud & Data	AWS, Azure, Snowflake, Metabase, environment validation, test-data orchestration
Programming	Python, C#, Java, TypeScript, SQL

Experience

Carta

Toronto, ON

SENIOR AUTOMATION ENGINEER

Oct 2024 - Present

- Drove shift-left quality for Fund Admin by translating high-risk **AMM**, **Payments**, and **KYC** requirements into automated **money movement** and compliance test coverage, improving early defect detection in release-critical systems.
- Improved CI/CD performance for the **Magni (Python/Selenium E2E framework)** by introducing PR-aware routing for test-only changes, reducing feedback time from 9–10 minutes to 1.5 minutes (85% reduction) while preserving full regression coverage for production-impacting changes.
- Built a **Feature Coverage Analysis** system mapping E2E execution traces to product surfaces via **Django** resolution and CI metadata, integrated into **Jenkins** and the Magni telemetry dashboard to surface coverage gaps and guide automated remediation.
- Developed a **Claude**-powered Magni plugin with multi-agent workflows for test generation, execution, debugging, and validation, enabling automated coverage gap closure and faster E2E test creation driven by coverage insights.
- Built AI-assisted engineering workflows using **Claude Skills** and **GitHub Actions** to automate dependency upgrades, PR creation, review assistance, and PR complexity analysis, improving developer throughput and reducing manual engineering overhead.
- Designed **Verdict** evaluation frameworks to validate correctness, reliability, and regression behavior of LLM-powered plugins, ensuring quality and stability of AI-driven test automation.

Telus Health (formerly LifeWorks)

Toronto, ON

SOFTWARE DEVELOPMENT ENGINEER IN TEST

Nov 2020 - Sep 2024

- Built and maintained **RESTful API** test automation using **Postman** and **Node.js**, supporting 30+ daily integration events between **Dynamics 365** and mobile platforms.
- Improved ERP application quality by 20% by developing a **.NET (C#)** automation framework with **SpecFlow**, accelerating functional and UAT feedback cycles.
- Integrated automated test execution into **CI/CD** pipelines using **Octopus Deploy** and **Azure DevOps**, enabling 50+ test runs per deployment and improving release confidence for data-heavy workflows.
- Enhanced release reliability by implementing **Elastic Reindexing** validation and automated **T-SQL**-driven verification scripts within deployment pipelines.
- Standardized QA knowledge and execution practices by maintaining end-to-end automation documentation in **Confluence**, improving team alignment and onboarding efficiency.

Cognitive Systems Corporation

Waterloo, ON

SOFTWARE ENGINEER - DSP ALGORITHMS (CO-OP)

Sep 2019 - Dec 2019

- Improved **WiFi motion** zone detection algorithms by refining signal-processing logic in **Python/NumPy**, increasing localization accuracy to sub-meter precision in test environments.
- Built and optimized simulation pipelines for WiFi link throughput degradation on **Linux**, reducing validation runtime by 30% and accelerating iteration cycles for algorithm tuning.
- Developed automated analysis scripts to compare predicted vs. observed motion events, improving debugging speed and reducing manual inspection effort during regression testing.
- Supported algorithm reliability by maintaining monitoring dashboards in **Grafana** and coordinating issue tracking and sprint execution using **Trello**.

Education

University of Waterloo

Waterloo, ON

BACHELOR OF APPLIED SCIENCE IN ELECTRICAL ENGINEERING

Sep 2015 - Jun 2020

- Algorithms and Data Structures, Database Systems, Control Systems, RF & Microwave Circuits, Management Science
- **Certificates:** Claude Fundamentals, Azure Fundamentals, Azure Developer Associate